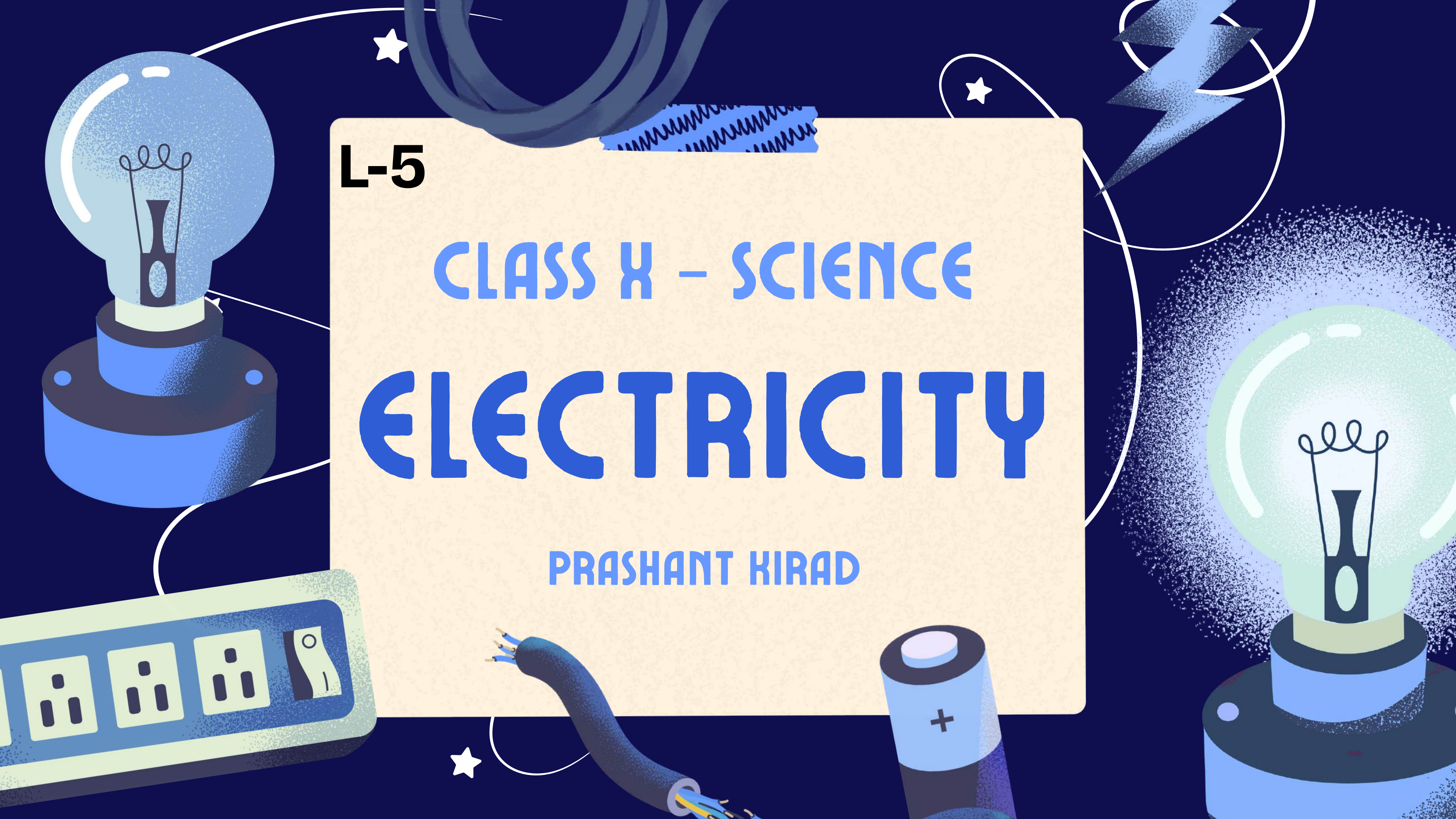


L-5

CLASS X – SCIENCE

ELECTRICITY

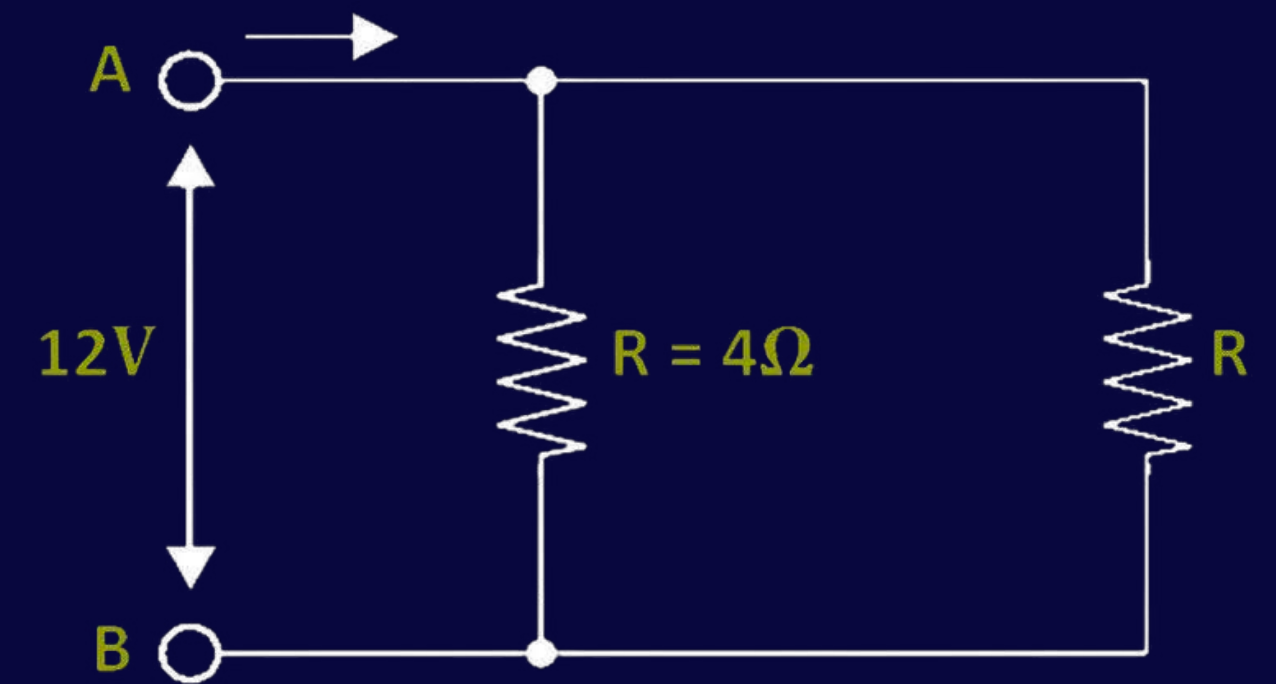
PRASHANT KIRAD



JOSH METER



Q. A student has two resistors of $2\ \Omega$ and $3\ \Omega$. She has to put one of them in place of R as shown in the circuit. The current needed in the entire circuit is exactly 9 A. Show by calculation which of the two resistors she should choose.



Q. Find out the following in the electric circuit given in the figure:

(a) Effective resistance of two $8\ \Omega$ resistors in the combination

(b) Current flowing through $4\ \Omega$ resistor

(c) Potential difference across $4\ \Omega$ resistance

(d) Difference in ammeter readings, if any.

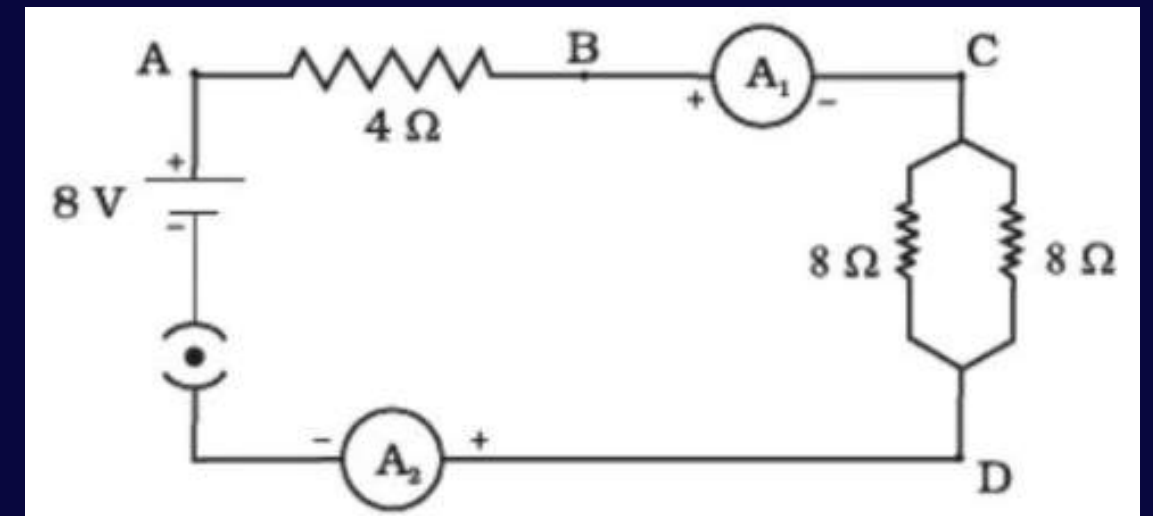
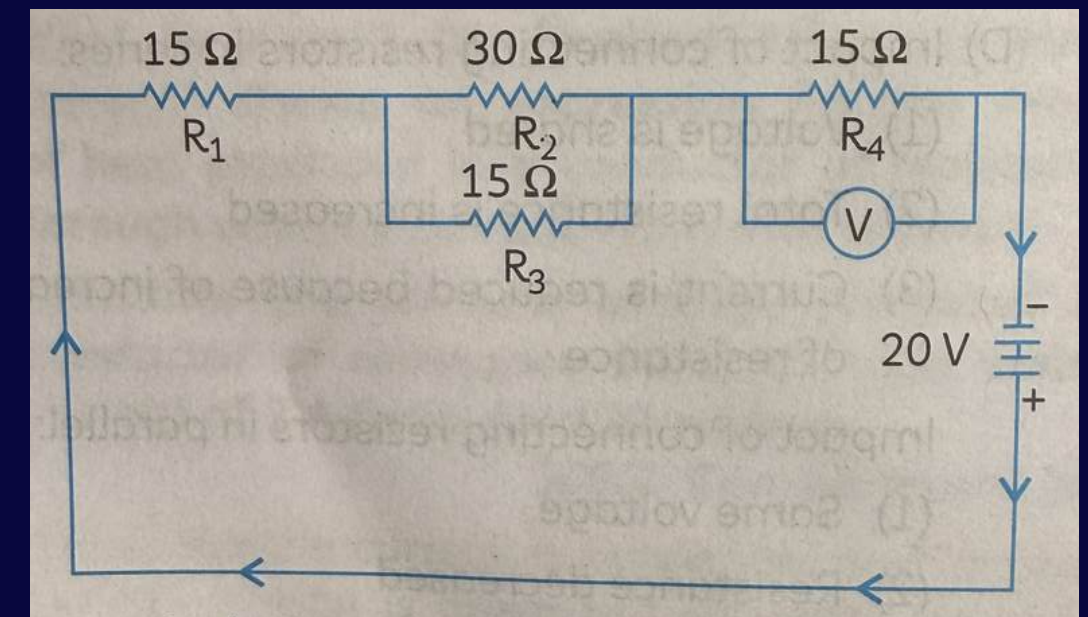


Fig. 12.9

Q. Four resistors, a voltmeter and a battery are connected in a circuit as shown below.

(A) What is the net resistance in the circuit?

(B) How much potential difference will the voltmeter connected across resistor R_4 measure?



Q. The circuit in the figure has resistors in series and parallel.

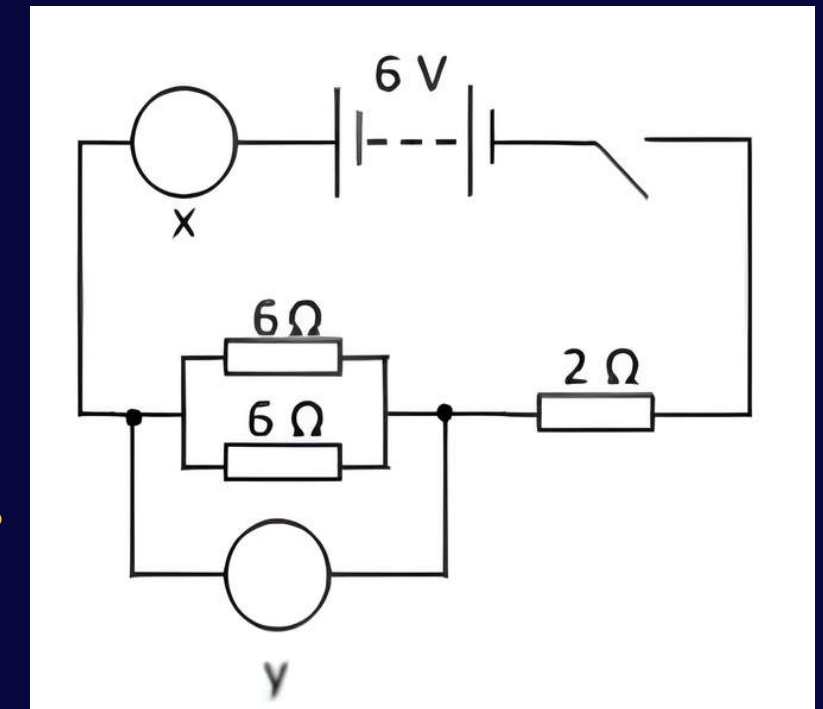
(A) State two factors that affect the resistance of a resistor.

(B) X and Y are meters. Identify X and Y and the electrical property that each meter measures.

(C) (i) Calculate the equivalent resistance in the circuit.

(ii) Using your answer, calculate the current in the circuit. State the unit.

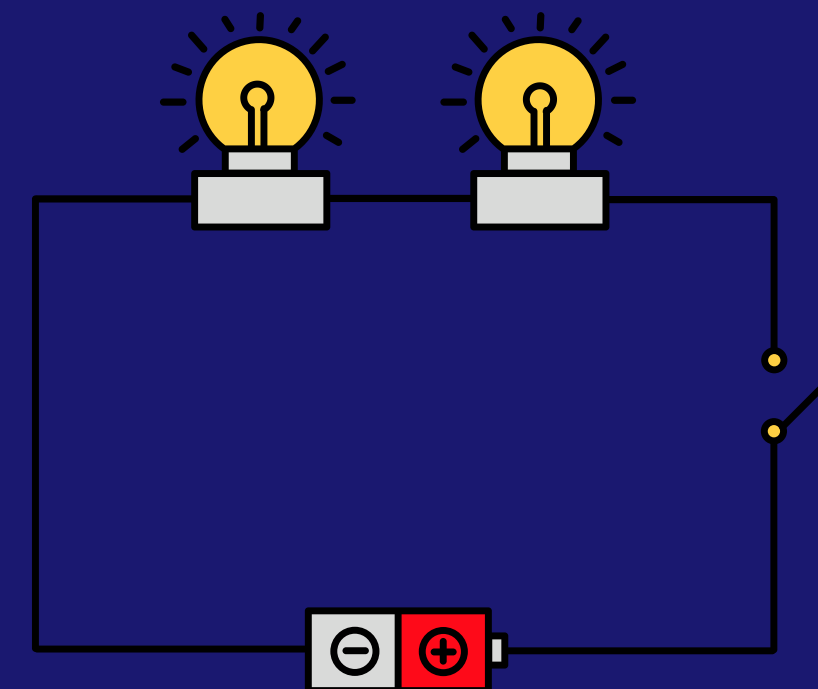
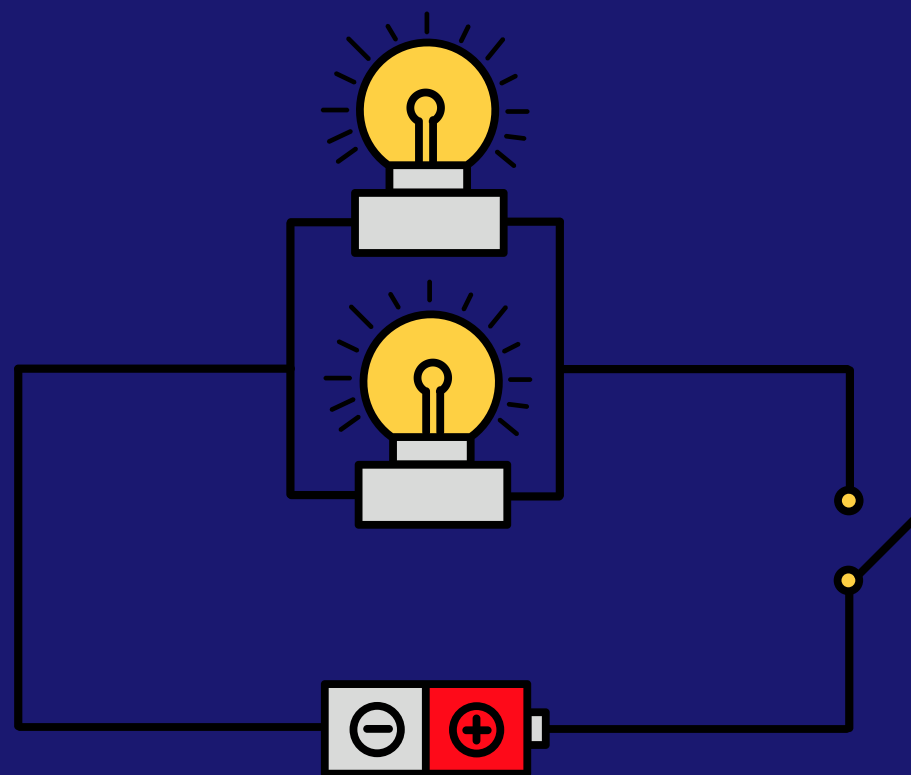
(D) Explain the impact of connecting resistors in series and connecting resistors in parallel on the resistance of a simple circuit.



What are the advantages of connecting electrical devices in parallel with the battery instead of connecting them in series? → NCERT

Advantage of parallel over series combination:

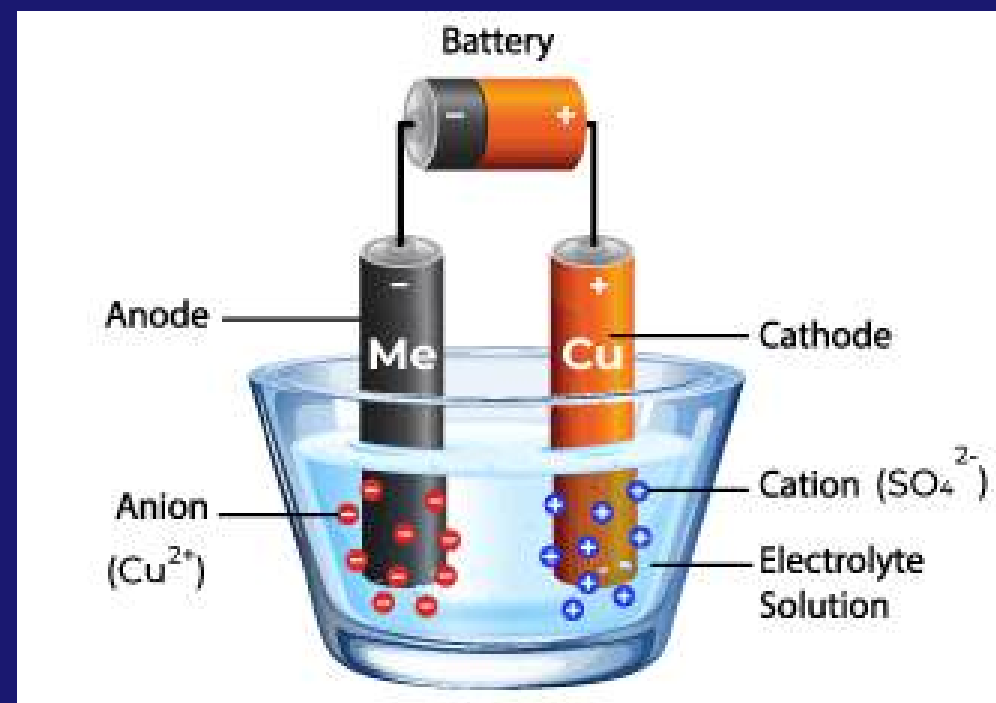
1. **Independent Functioning:** Each device works independently. If one fails, others continue to work.
2. **Same Voltage:** All components receive the same voltage as the source.
3. **Reduced Resistance:** Adding more devices decreases the total resistance, allowing efficient current flow.
4. **Easy to Add Devices:** New devices can be added without affecting others.



THREE EFFECTS OF CURRENT

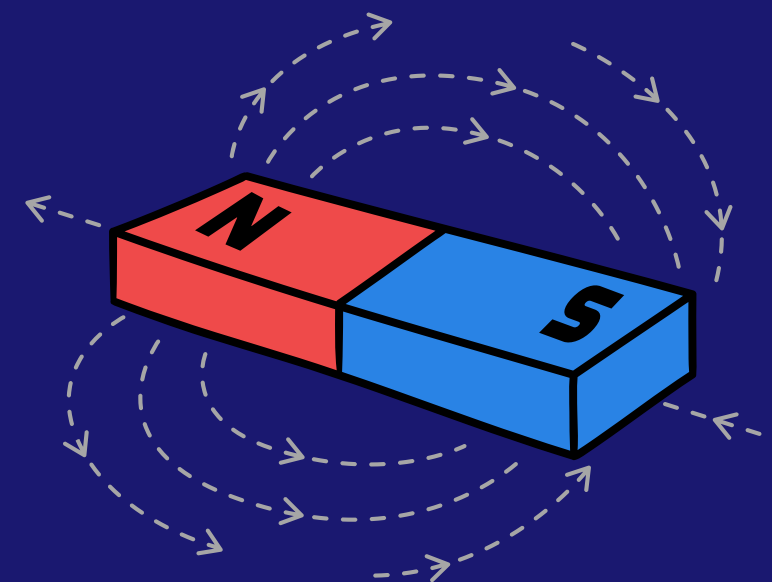
Chemical Effects

- When an electric current flows through the conducting solution, it triggers chemical reactions to take place at the cathode and anode.
- Example** - Electroplating, Hydrolysis, Formation of Sodium Hydroxide (NaOH)



Magnetic Effects

When an electric current flows through a conductor, a magnetic field is generated in the surrounding area.



Heating Effects

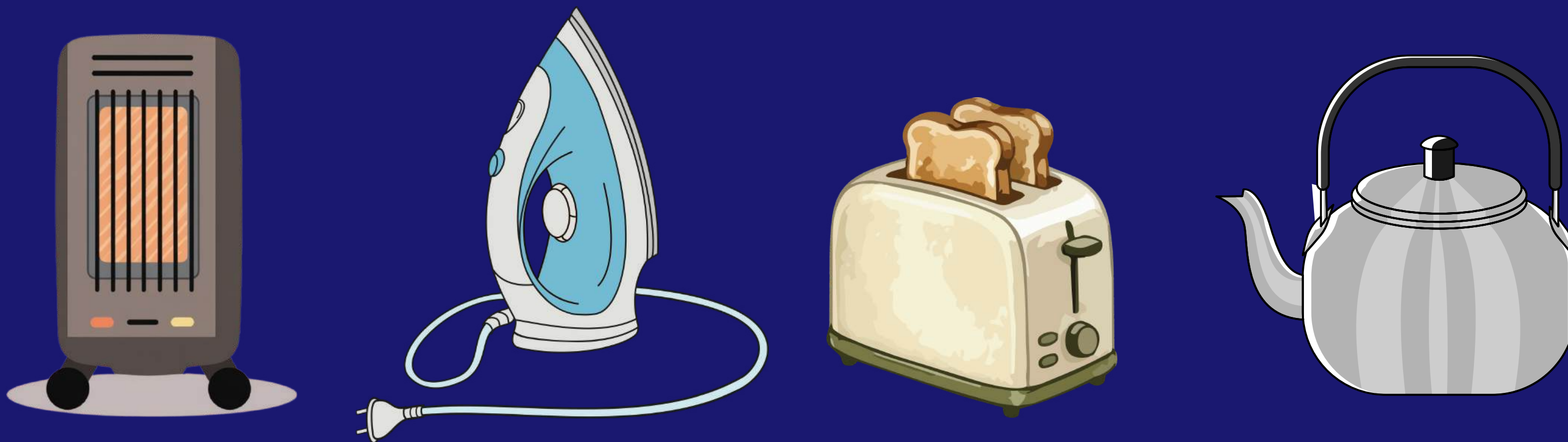
When an electric current flows through a conductor, heat energy and power are dissipated as a result of repeated collisions between moving charges and atoms.



HEATING EFFECT OF ELECTRIC CURRENT

When electric current passes through a resistor, **electrical energy is converted into heat energy**. This is called the heating effect of electric current.

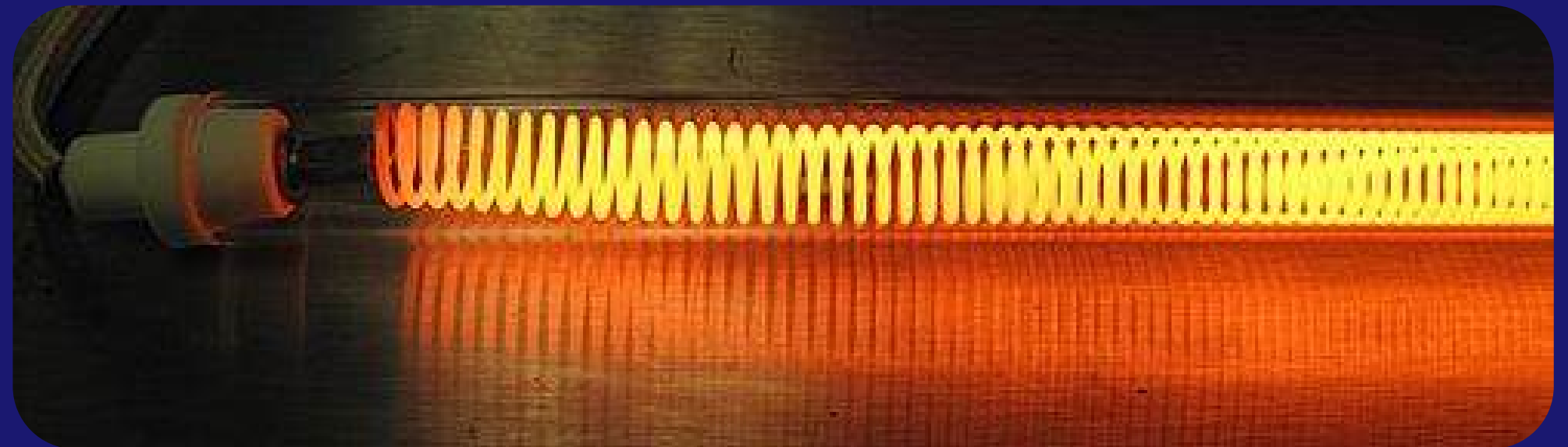
Examples: Electric heater, electric iron, toaster, kettle, etc.



JOULE'S LAW OF HEATING

It states that, *“the heat generated in a conductor is directly linked to the square of the current (I), the resistance (R) of the conductor, and the time period (T) of the current flow.”*

$$H = I^2 R t$$



$$V = \frac{W}{Q} \rightarrow W = VQ \quad \text{--- (1)}$$

as well as $I = \frac{Q}{t} \quad \text{--- (2)}$

② in ①

$$W = VIt \quad \leftarrow V = IR$$

$$W = I^2 R t$$

// $H = I^2 R t$

[NCERT]

Q. 100 J of heat is produced each second in a $4\ \Omega$ resistance. Find the potential difference across the resistor.

[NCERT]

Q. Compute the heat generated while transferring 96000 coulomb of charge in one hour through a potential difference of 50 V.

[NCERT]

Q. An electric iron of resistance $20\ \Omega$ takes a current of $5\ \text{A}$. Calculate the heat developed in $30\ \text{s}$.



Q. A complete circuit is left on for several hours, causing the connecting copper wire to heat. As the temperature of the wire increases, the electrical resistance of the wire:

- (A) Decreases**
 - (B) Remains the same**
 - (C) Increases**
 - (D) Increases for some time and then decreases**
- 

Q. If the current passing through a heater is halved, then the heat produced by it becomes:

A. $H/2$

B. $4H$

C. $H/4$

D. $2H$

PRACTICAL APPLICATIONS OF HEATING EFFECTS

Electric Bulb:

In an electric bulb, the heating effect is used to produce light. The filament of the bulb becomes very hot due to electric current and starts glowing.

Why is tungsten used in bulb filament?

Tungsten is used because:

1. It has a very high melting point.
2. It can become very hot without melting.
3. It emits light when heated strongly.

The bulb is filled with **nitrogen and argon** gases to increase the life of the filament.



Electric Fuse:

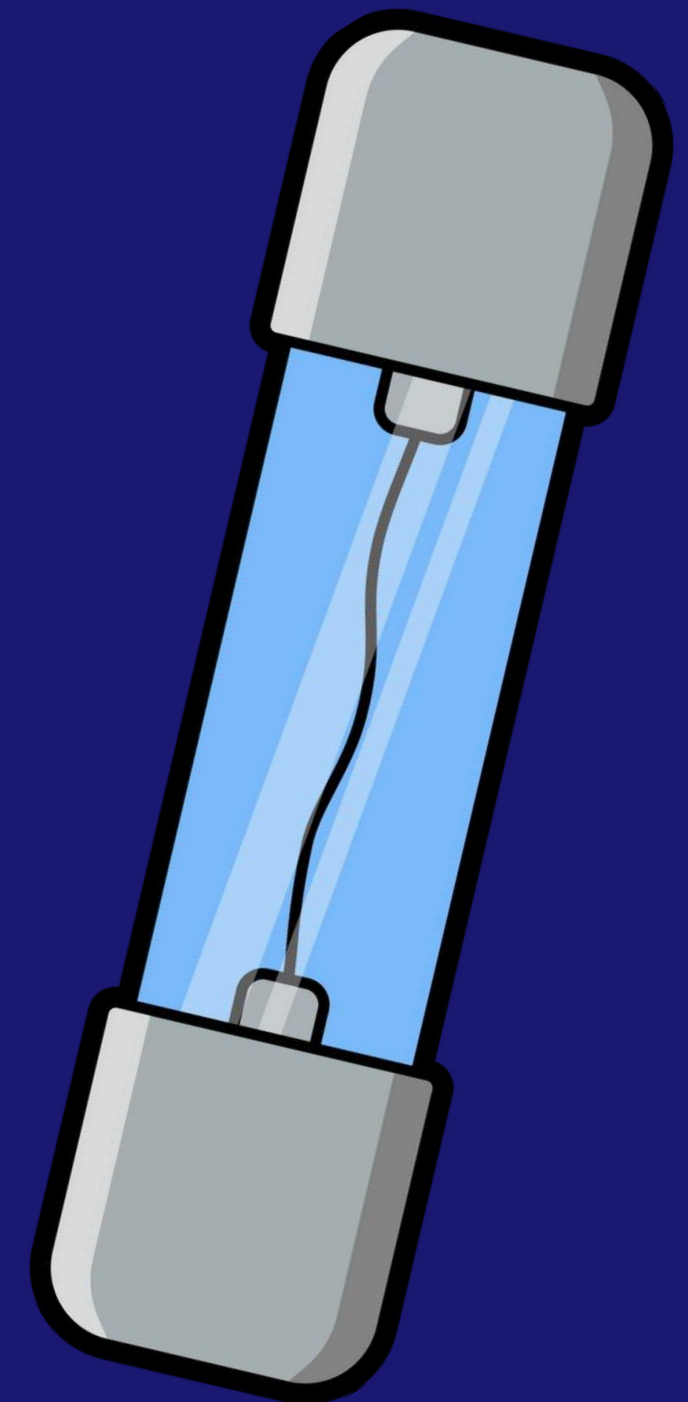
A fuse is a safety device used in electric circuits. It protects electrical appliances from very high current.

Working of Fuse

- A fuse wire is connected in series with the appliance.
- When current becomes larger than the safe value, the fuse wire gets heated and melts.
- This breaks the circuit and stops the flow of current.
- Thus, the appliance is protected from damage.

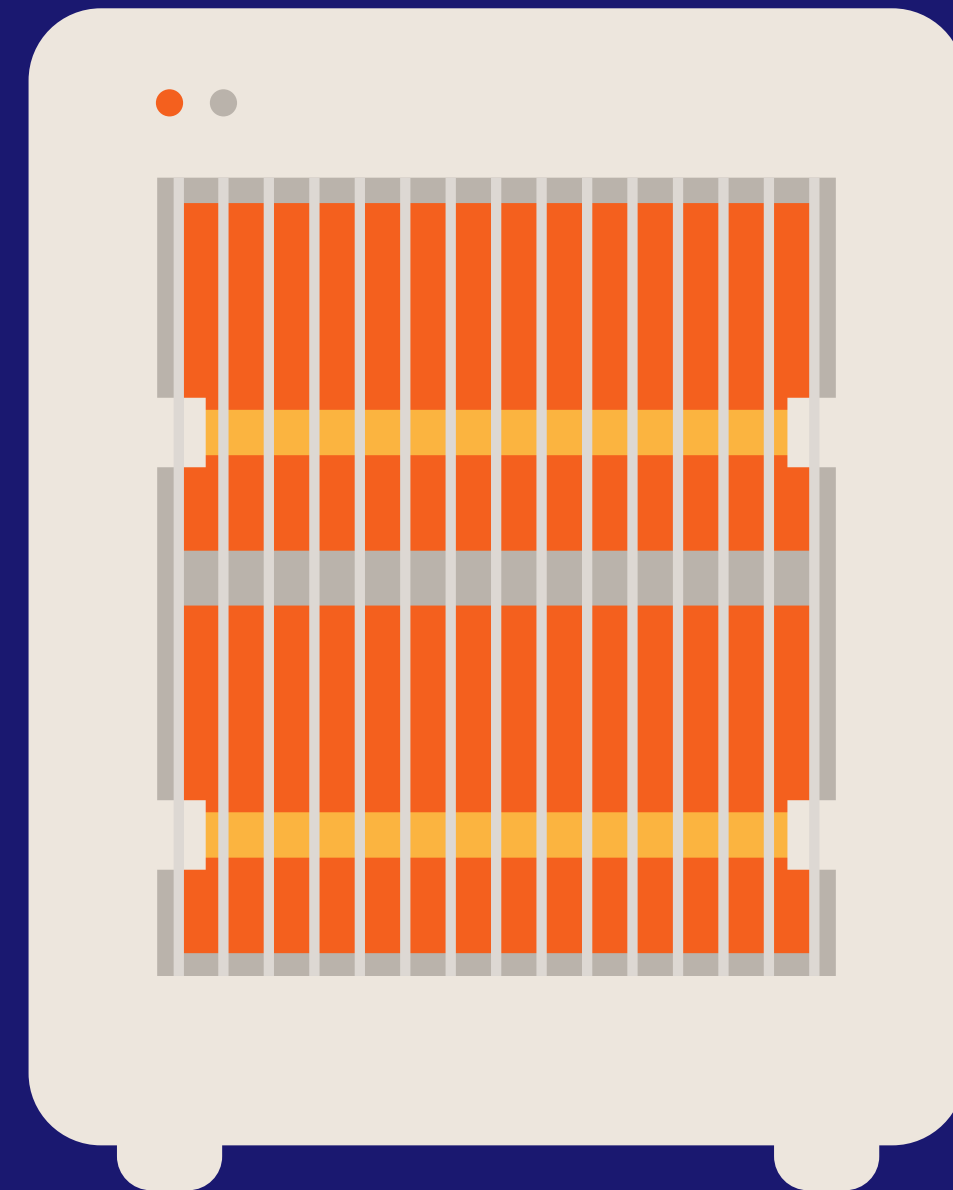
Fuse Ratings

Domestic fuses are commonly rated as: 1 A, 2 A, 3 A, 5 A, 10 A



Electric Heater/ Heating element:

- In an electric heater, a high-resistance **nichrome wire** is used as the heating element.
- This nichrome wire is made into a coil and placed in grooves made of ceramic material or china clay, because these materials can withstand high temperature.
- When electric current flows through the nichrome coil, it becomes very hot due to the heating effect of electric current. This heat is then used for heating purposes, such as heating cooking vessels.



Elements used :

- **Filament of the bulb - Tungsten**
- **Connecting wires - Copper**
- **Heating Elements - Nichrome**
- **Fuse wire - Sn - Pb alloy**

The image features a motivational quote centered over a background of a sunset or sunrise over a body of water. The sun is low on the horizon, creating a bright, glowing effect that reflects on the water. The sky is a mix of orange, yellow, and light blue. The water is dark with some ripples and small, light-colored patches. The quote is written in a bold, black, sans-serif font, arranged in five lines. The overall mood is inspirational and hopeful.

**THINK BIG.
TRUST
YOURSELF
AND MAKE
IT HAPPEN**